

Image Classification Extension

<https://github.com/aiyshwary/Image-Classification-Extension-in-Rapid-Miner>

Python

PyTorch

ONNX

RapidMiner

Transfer Learning

OVERVIEW

A RapidMiner Studio extension enabling no-code transfer learning for image classification, supporting fine-tuning of popular CNN architectures on custom datasets and ONNX-based inference without writing a single line of code.

IMPACT

Developed a RapidMiner extension with two custom operators — Fine Tuning and Inference — enabling practitioners to apply transfer learning with ResNet, VGGNet, DenseNet, Inception and more using a drag-and-drop interface.

TECHNICAL WALKTHROUGH

1. Image Classification Extension

Problem Statement In real-world scenarios, image datasets vary significantly in size, quality, and domain. A fixed model often fails to generalize well. Hence, there is a need for a customizable image classification system that can adapt to different datasets while maintaining good accuracy.

Approach Built an image classification extension using TensorFlow and PyTorch Leveraged transfer

learning with pre-trained models such as: ResNet AlexNet VGG SqueezeNet DenseNet Inception Vision Transformers (ViT, DeiT via Hugging Face) Fine-Tuning Dataset organized in class-wise folders

Automatic 75% training / 25% validation split Customizable parameters: Number of layers Batch size

Learning rate Optimizer Epochs Used data augmentation, class-weighted learning, and callbacks like early stopping and learning-rate scheduling Inference Trained model exported to ONNX Images are

preprocessed and passed to the model Output includes: Predicted class label Confidence score Images

are automatically placed into output folders based on predicted class Results Animal classification: ~87% accuracy Facial emotion recognition: ~80% accuracy Flower dataset: ~94% accuracy

KEY DETAILS

- Fine Tuning operator accepts an image directory and hyperparameters; outputs a fine-tuned model in ONNX format.
- Inference operator loads the ONNX model and outputs a prediction DataFrame with classified labels.
- Supports ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, and Inception architectures.
- Compatible with both CPU and GPU (CUDA) environments via configurable conda environment.
- Installed as a JAR extension in RapidMiner Studio with Python Scripting backend.